

Project Summary

QMatBridge: Open Cyberinfrastructure for Reproducible First-Quantized Hamiltonian Simulation Benchmarks

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Overview

Research groups developing fault-tolerant Hamiltonian simulation algorithms — block-encodings, qubitization circuits, linear-combination-of-unitaries (LCU) decompositions — require realistic electronic-structure Hamiltonians to test and benchmark their primitives against physically meaningful targets. The standard path today is a collection of private, incompatible scripts that pull density functional theory (DFT) data from databases such as the Materials Project [1] or OQMD [2], reformat the data, and feed it into a quantum compiler. These scripts are almost never shared, rarely interoperable with more than one quantum framework, and frequently omit the provenance information required to reproduce or compare results across institutions. When two groups report T-gate counts for the same material, there is no current mechanism to verify they used the same Hamiltonian, the same basis-set parameters, or the same LCU decomposition conventions.

QMatBridge addresses this structural gap by providing an open-source, dependency-free neutral intermediate representation (NIR) that any upstream database adapter can write to and any downstream quantum resource estimator or compiler can read from, with full provenance embedded at the record level. This proposal requests support to (i) complete the QMatBridge v0.2–v0.5 development roadmap, including live Materials Project and OPTIMADE adapters; (ii) build and curate a Tier-1 benchmark fixture set spanning five representative crystalline materials; (iii) implement exporters for OpenFermion [3], Qualtran [4], and pyLIQTR [5]; and (iv) establish a community benchmark standard through open governance and a dedicated sensitivity-analysis framework for resource-estimate parameter studies.

Intellectual Merit

The proposed work advances the computational and data-centric methodology of materials research in three ways. First, a *canonical hash* (SHA-256 over physically meaningful fields of a `QMatEntry` record) provides a unique, database-independent identity for each Hamiltonian, enabling benchmark reproducibility at the record level and directly addressing the cross-code reproducibility crisis documented in DFT [6]. Second, a *reproducible sensitivity analysis framework* — demonstrated in preliminary results — identifies which input controls (plane-wave cutoff energy, energy precision target $\delta\epsilon$, and orbital band count n_b) govern estimated T-gate and qubit resource costs; the analysis reveals that $\delta\epsilon$ dominates with near-unit inverse elasticity ($\epsilon \approx -1.01$), providing a quantitative planning heuristic with direct implications for benchmark selection and hardware targeting. Third, the QMatBridge *oracle metadata layer* stores LCU 1-norms, SELECT/PREPARE circuit parameters, and complexity annotations in a structured, queryable format, enabling systematic multi-material Pareto analysis across competing resource objectives and supporting the algorithm–materials co-design cycle.

Broader Impacts

QMatBridge directly addresses a community infrastructure deficit affecting every research group at the interface of computational materials science and fault-tolerant quantum algorithms. All outputs are MIT-licensed with an open governance model. Situated at Northwestern University's MSE department — with existing connections to the Argonne National Laboratory quantum computing program and the NSF-funded Institute for Quantum Information Research and Engineering (INQUIRE) at Northwestern — the project bridges the DFT computation community and the quantum algorithms community, providing training opportunities for graduate students at this increasingly demanded interface. The benchmark suite is designed for integration into graduate courses on quantum algorithms and materials informatics, and the CDS&E submission route reflects the project's explicit commitment to software sharing, FAIR data principles [7], and long-term community maintainability.

Project Description

QMatBridge: Open Cyberinfrastructure for Reproducible First-Quantized Hamiltonian Simulation Benchmarks

Introduction and Motivation

Fault-tolerant quantum simulation of electronic structure holds the potential to resolve long-standing challenges in materials science — including the accurate treatment of strongly correlated electron systems, prediction of high-temperature superconductors, and design of transition-metal catalysts — that remain intractable for classical computation at the required accuracy [8–10]. The theoretical framework for these simulations is now mature: first-quantized plane-wave Hamiltonians combined with qubitization or LCU decompositions provide asymptotically optimal fault-tolerant algorithms for periodic materials [11–13], with well-characterized T-gate and qubit resource estimates scaling as $\mathcal{O}(\lambda/\delta\varepsilon)$ and $\mathcal{O}(\log N_{\text{PW}})$, respectively, where λ is the LCU 1-norm, $\delta\varepsilon$ is the target energy precision, and N_{PW} is the number of plane-wave basis functions.

Despite this theoretical maturity, the field lacks a shared data infrastructure connecting classical DFT databases to these quantum algorithms. A researcher wishing to benchmark a new LCU circuit variant against bulk silicon must currently: (1) access the Materials Project API [1] and extract structure, basis-set parameters, and pseudopotential metadata; (2) reformat this data into the conventions of a specific quantum compiler; (3) compute the LCU 1-norm λ according to the conventions of their chosen decomposition; and (4) record sufficient metadata for others to reproduce the computation. Each step is currently performed with private, group-specific scripts that are incompatible across groups. The consequence, documented in analogy with the DFT reproducibility crisis [6], is that T-gate estimates for the same physical system vary substantially across publications — not due to algorithmic differences but due to inconsistent Hamiltonian parameterization.

QMatBridge resolves this by providing an open-source, zero-dependency neutral intermediate representation (NIR) for first-quantized electronic Hamiltonians. The framework defines a `QMatEntry` schema that records the full provenance chain from upstream database to oracle metadata, exposes a deterministic canonical hash for record identity, and is designed to be targeted by any database adapter and consumed by any quantum compiler or resource estimator. This proposal requests CDS&E support to complete the adapter and exporter layers, build a Tier-1 community benchmark fixture set, and establish the sensitivity analysis framework as a reproducible methodology for parameter studies in quantum simulation planning.

New science enabled. QMatBridge is not a convenience tool; it makes scientifically distinct studies possible. Three classes of investigation that are currently impossible without it: (1) *Cross-group λ reproducibility studies*: by providing a canonical hash, two independent groups can verify for the first time that their T-gate estimates are based on the same Hamiltonian, enabling the first systematic meta-analysis of resource estimation uncertainty across the literature. (2) *Precision-target optimization*: the sensitivity analysis framework allows systematic identification of the $\delta\varepsilon$ threshold below which resource costs increase super-linearly for each material class, informing hardware roadmap targeting in a way not previously quantified. (3) *Multi-functional consistency studies*: by supporting PBE, PBEsol, HSE06, and r²SCAN via the Alexandria adapter, QMatBridge enables quantification of how functional choice shifts λ for strongly correlated materials, directly informing

which materials require HSE06-level accuracy for reliable resource estimates.

Community validation of the gap. The need for reproducible quantum simulation benchmark infrastructure is independently recognized across the field, though existing efforts address adjacent rather than overlapping problem domains. Clark et al. (NQAC, 2026) [14] are developing an open-source, executable benchmark framework for *near-term quantum chemistry algorithms on molecular systems* (second-quantized, small molecules, NISQ hardware). QREChem [15] provides logical resource estimation for *molecular* ground-state energy calculations via Trotter-based QPE. ArQTiC [16] enables near-term simulation of *materials governed by one-dimensional Heisenberg Hamiltonians* (real- and imaginary-time evolution, near-term hardware). None of these tools addresses *periodic crystalline solids* in the *first-quantized, fault-tolerant* regime — the regime in which quantum advantage over classical DFT is expected to emerge for industrially relevant bulk materials [9, 17]. QMatBridge is designed precisely for this unoccupied niche. Rather than competing with Clark et al., QREChem, or ArQTiC, QMatBridge is the *materials-and-fault-tolerant counterpart* to these molecular and near-term tools: the four together cover the full spectrum from small molecules on NISQ hardware to periodic crystals on fault-tolerant hardware.

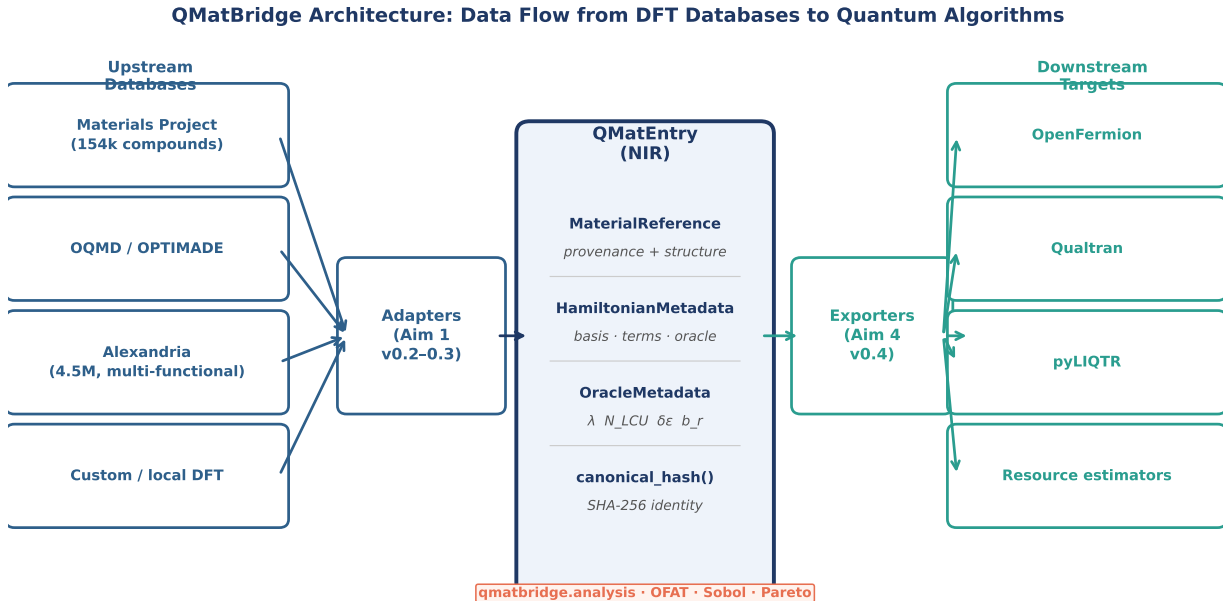


Figure 1: QMatBridge architecture. Adapters (Aim 1) translate upstream database records into a versioned, hash-addressed `QMatEntry` neutral intermediate representation (NIR). Exporters (Aim 4) translate the NIR into formats consumed by downstream quantum compilers and resource estimators. The `qmatbridge.analysis` module (Aim 3) operates on libraries of `QMatEntry` records to produce sensitivity analyses and Pareto frontier rankings. All components are dependency-free at the core schema level.

Background

First-Quantized Hamiltonian Simulation

In the first-quantized plane-wave formalism [11, 12], the electronic Hamiltonian of a periodic crystalline material is partitioned as $H = T + U + V$, where T is the kinetic energy operator

(diagonal in momentum space), U is the electron–electron Coulomb interaction, and V is the electron–nuclear interaction. The LCU decomposition expresses $H = \sum_{\ell} \alpha_{\ell} H_{\ell}$, where the H_{ℓ} are unitary operators and $\alpha_{\ell} \geq 0$ are real coefficients. The total LCU 1-norm $\lambda = \sum_{\ell} \alpha_{\ell}$ controls the dominant cost of qubitized quantum phase estimation (QPE) [18]: the number of Toffoli gates scales as $\mathcal{O}(\lambda/\delta\varepsilon)$, where $\delta\varepsilon$ is the target energy precision. For bulk silicon at a 520 eV plane-wave cutoff, representative values are $\lambda \approx 316$ Ha and $N_{\text{PW}} \approx 1849$ plane waves per spin channel, yielding a state register of $\lceil \log_2 N_{\text{PW}} \rceil = 11$ qubits [11].

This formalism is the basis for a growing body of material-specific resource estimates [8–10, 17], but the underlying Hamiltonians are not shared, making cross-study comparison unreliable. The QMatBridge `OracleMetadata` schema is designed to store precisely the quantities needed for reproducible qubitization resource estimation: λ , N_{LCU} , $\delta\varepsilon$, η (phase-error target), $\lceil \log_2 N_{\text{PW}} \rceil$, and the number of rotation bits b_r , along with per-term breakdowns for T , U , and V .

The Reproducibility Gap

The materials DFT community confronted an analogous reproducibility crisis in 2016, when Lejaeghere et al. [6] compared 40 DFT codes computing equations of state for 71 elemental solids and found systematic deviations attributable to basis-set conventions, pseudopotential families, and code-specific defaults. The resolution required not algorithmic innovation but infrastructure: standardized input files, public reference data sets, and cross-code validation workflows.

The quantum simulation community faces a structurally identical problem, with added complexity: the connection between DFT outputs and quantum algorithm inputs involves non-trivial transformations (plane-wave count computation, LCU norm assembly, rotation-bit sizing) that are currently performed differently in each group’s codebase. QMatBridge addresses this at the infrastructure level by defining the transformation chain in a single, public, versioned schema and implementing it in a reference codebase subject to continuous integration testing.

Existing Software Landscape and Innovation Gap

Table 1 positions QMatBridge against the current software ecosystem. No existing tool simultaneously provides database-agnostic provenance tracking, a canonical Hamiltonian identity, and direct export to multiple fault-tolerant quantum compilers.

QMatBridge’s two most distinctive innovations are (i) the *canonical hash*, which for the first time gives fault-tolerant quantum simulation benchmarks a database-independent, tamper-evident identity analogous to a DOI for a specific Hamiltonian, and (ii) the *sensitivity analysis framework*, which transforms resource estimation from a single-point calculation into a reproducible parameter study, enabling systematic reasoning about where quantum advantage thresholds are most sensitive to computational choices.

The adjacent tools ArQTIC [16], QREChem [15], and the recently funded Clark et al. benchmarking project [14] address complementary but distinct problem spaces: near-term hardware, second-quantized molecular Hamiltonians, or one-dimensional lattice models. Their existence independently validates the community’s recognition that benchmark infrastructure is a field-wide need. QMatBridge fills the remaining gap: *periodic crystalline materials, first-quantized plane-wave basis, fault-tolerant algorithms, and DFT database provenance* — the combination required for practical quantum advantage in bulk materials simulation [9, 17, 19]. None of the existing or adjacent tools is architecturally positioned to fill this niche, because all of them begin after the Hamiltonian is already specified and are not connected to the upstream DFT database ecosystem.

Table 1: Comparison of QMatBridge with existing software at the DFT–quantum algorithm interface. ✓ = supported; — = not supported; ~ = partial.

Feature	OpenFermion	Qualtran	pyLIQTR	AiiDA	pymatgen	QMatBridge
DFT database adapters	—	—	—	~	~	✓
Canonical Hamiltonian hash	—	—	—	—	—	✓
First-quantized LCU schema	—	~	~	—	—	✓
Multi-compiler export	~	—	—	—	—	✓
Sensitivity / Pareto tools	—	—	—	—	—	✓
Zero-dependency core	✓	—	—	—	—	✓
Community benchmark fixtures	—	—	—	—	—	✓
<i>Scope comparison with adjacent near-term / molecular tools:</i>						
Periodic cryst. (fault-tolerant)	—	~	~	—	—	✓
Molecular (NISQ / near-term)	✓	~	~	—	—	—
Provenance from DFT database	—	—	—	~	~	✓

Adjacent tools not shown above (different scope): ArQTiC [16] (near-term, 1D Heisenberg materials); QREChem [15] (fault-tolerant resource estimation, molecular); Clark et al. [14] (near-term benchmark framework, molecular). None of the three targets periodic crystalline solids with first-quantized fault-tolerant algorithms or DFT database provenance.

Preliminary Results

The QMatBridge v0.1 codebase [20] demonstrates the feasibility of the proposed approach across four independent dimensions.

Schema and Canonical Hash (v0.1)

The `QMatEntry` schema is implemented as a dependency-free Python dataclass hierarchy comprising `MaterialReference` (provenance + structure), `HamiltonianMetadata` (basis, terms, oracle), and `OracleMetadata` (LCU decomposition parameters). The `canonical_hash()` method computes a SHA-256 digest over 12 physically meaningful fields (upstream source, functional, formula, spacegroup, electron count, spin polarization, basis type, cutoff energy, band count, and oracle precision target), providing a stable identity that is invariant to metadata changes but sensitive to any physically relevant modification. A concrete silicon entry (mp-149, PBE/PAW_PBE/VASP, 520 eV cutoff, $\lambda = 315.8$ Ha, $\delta\varepsilon = 10^{-3}$ Ha) has been implemented and validated against the published resource estimates of Babbush et al. [11], confirming that the schema captures all quantities required to reproduce the literature estimate.

Sensitivity Analysis of Resource Estimates

A one-factor-at-a-time (OFAT) sensitivity analysis was performed over three principal input controls of the silicon-like baseline: $E_{\text{cut}} \in \{400, 450, 500, 550, 600\}$ eV, $\delta\varepsilon \in \{5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$ Ha, and $n_b \in \{12, 16, 20, 24\}$ bands. Proxy models for the sweep are:

$$N_{\text{PW}} = N_{\text{PW}}^{(0)} \left(\frac{E_{\text{cut}}}{E_{\text{cut}}^{(0)}} \right)^{3/2}, \quad n_{\text{state}} = \lceil \log_2 N_{\text{PW}} \rceil, \quad n_{\text{rot}} = \lceil -\log_2 \delta\varepsilon \rceil + 10, \quad (1)$$

$$\lambda = \lambda^{(0)} \left(\frac{E_{\text{cut}}}{E_{\text{cut}}^{(0)}} \right)^{1/2} \left(\frac{n_b}{n_b^{(0)}} \right)^{1/4}, \quad (2)$$

$$\tilde{T} = \lambda / \delta\varepsilon, \quad q_{\text{prox}} = n_{\text{state}} + n_{\text{rot}}, \quad \mathcal{S} = \tilde{T}(1 + q_{\text{prox}}/100), \quad (3)$$

where superscript (0) denotes baseline values ($\lambda^{(0)} = 315.8$ Ha, $E_{\text{cut}}^{(0)} = 520$ eV, $N_{\text{PW}}^{(0)} = 1849$, $n_b^{(0)} = 16$). The formulas in Eqs. (1)–(3) are explicit planning proxies, not material-specific resource estimates: the λ exponents (0.5 in E_{cut} , 0.25 in n_b) are motivated by reciprocal-space volume scaling arguments and should be replaced by computed values from the Babbush et al. formula [11] once the v0.2 adapter is live (Aim 2). All exponents and constants are documented in the output JSON metadata for independent verification.

The resulting tornado impact ranking (Table 2) reveals that the energy precision target $\delta\varepsilon$ is the dominant control by a substantial margin, with 151.9% relative impact and a local arc elasticity $\varepsilon_{\delta\varepsilon} \approx -1.01$ — confirming near-unit inverse proportionality between the composite resource score and $\delta\varepsilon$. The cutoff energy and band count exhibit sublinear effects, with elasticities $\varepsilon_{E_{\text{cut}}} \approx +0.50$ and $\varepsilon_{n_b} \approx +0.25$, respectively, consistent with the power-law exponents in Eq. (2).

Table 2: Tornado impact summary from one-factor-at-a-time sensitivity analysis. Baseline composite score $\mathcal{S}_0 = 413,698$. All outputs are machine-readable in `sensitivity_tornado.csv` and `sensitivity_grid.json`.

Parameter	Range	$\mathcal{S}_{\min}$	$\mathcal{S}_{\max}$	Rel. impact	Elasticity
$\delta\varepsilon$ (Ha)	$[5 \times 10^{-4}, 2 \times 10^{-3}]$	205,270	833,712	151.9%	-1.01
E_{cut} (eV)	[400, 600]	362,837	447,775	20.5%	+0.50
n_b	[12, 24]	384,989	457,832	17.6%	+0.25

These findings carry direct practical implications: relaxing the precision target from $\delta\varepsilon = 5 \times 10^{-4}$ Ha (sub-chemical accuracy) to $\delta\varepsilon = 2 \times 10^{-3}$ Ha (approximately 1 kcal/mol, chemical accuracy) reduces the composite resource score by $\sim 75\%$, while reducing E_{cut} from 600 to 400 eV reduces it by only $\sim 12\%$. This quantitative hierarchy informs a planning heuristic that is currently absent from the field: *fix the precision target first, then optimize the basis-set parameters*.

Multi-Objective Pareto Analysis

A complementary Pareto frontier analysis was performed over five representative materials (Si, LiH, MgO, AlN, GaN) simultaneously minimizing the T-gate proxy $\tilde{T}$ and qubit proxy q_{prox} . The non-dominated set (identified by a pairwise dominance check) provides a rigorous benchmark selection criterion for studies constrained by hardware qubit count or coherence time. The frontier data are exported as machine-readable CSV and JSON artifacts with full provenance metadata.

Reproducibility Infrastructure

All examples are fully deterministic (fixed parameter ordering, no random sampling) and produce identical numerical output across platforms. Each script accepts a single command from the repository root (`python examples/script.py`) and generates CSV, JSON, and PNG artifacts that constitute citable computational data products. The canonical hash of the silicon baseline entry remains stable across schema patch releases, confirmed by a CI regression fixture.

Research Plan

The proposed work is organized into five research aims spanning three years. Aims 1–2 constitute the data infrastructure layer; Aims 3–4 implement the algorithm interface; Aim 5 establishes the community benchmark standard.

Aim 1 (Year 1): Complete database adapter layer — Materials Project (v0.2), OPTIMADE (v0.3), and Alexandria (v0.3).

Aim 2 (Year 1): Build and validate the Tier-1 benchmark fixture set for five canonical materials.

Aim 3 (Years 1–2): Implement the sensitivity analysis and resource estimation framework as a reusable library module.

Aim 4 (Year 2): Implement downstream exporters for OpenFermion, Qualtran, and pyLIQTR.

Aim 5 (Years 2–3): Establish community benchmark standard — governance, documentation, JOSS publication, and workshop.

Aim 1: Database Adapter Layer

Materials Project Adapter (v0.2)

The Materials Project [1] hosts > 154,000 inorganic compounds computed at the PBE/PAW level using VASP [21]. The `materials_project` adapter stub in QMatBridge v0.1 will be promoted to a live implementation using the `mp-api` client. Specifically, we will implement:

- `fetch_structure_metadata_from_mp(mp_id)`: retrieves lattice parameters, site positions, species, and spacegroup from the Materials Project REST API and populates a `StructureMetadata` and `LatticeMetadata` record.
- `fetch_hamiltonian_metadata_from_mp(mp_id, ecut_ev)`: retrieves VASP ENCUT from the task document, k-point mesh from the `kpoints` sub-document, and pseudopotential family from `potcar_symbols`; populates `BasisMetadata` and `SourceProvenance`.
- `num_plane_waves_from_ecut(ecut_ev, volume_ang3)`: computes N_{PW} via the reciprocal-space volume relation $N_{PW} = (4\pi/3)(2mE_{cut}/\hbar^2)^{3/2}\Omega/(2\pi)^3$, where Ω is the unit-cell volume. This utility will be unit-tested against the literature value for mp-149 ($N_{PW} = 1849$ at 520 eV).

Integration tests, marked `@pytest.mark.integration` and skipped when `MP_API_KEY` is absent, will verify the silicon round-trip: adapter fetch $\rightarrow$ `QMatEntry` construction $\rightarrow$ `canonical_hash()` comparison against the committed fixture.

Risk mitigation for Aim 1a: If the MP API undergoes breaking changes, the adapter will be pinned to a stable API version and the fallback path uses cached task documents committed as test fixtures. The OPTIMADE adapter (Aim 1b) independently covers AFLOW, JARVIS, and Materials Cloud, so Aim 2 is not exclusively dependent on MP access.

OPTIMADE and Alexandria Adapters (v0.3)

The OPTIMADE standard [22] provides a unified REST API covering AFLOW, JARVIS, NOMAD [23], and Materials Cloud, collectively spanning > 10^7 DFT-computed structures. A single `OptimadeAdapter` class will be implemented using the `/structures/{id}` endpoint, with provider-specific URL configurations for each database. The canonical source string will encode both the protocol and provider (e.g., "optimade:aflow") to ensure cross-database deduplication via shared ICSD numbers where available.

The Alexandria database [24], providing ~ 4.5 million PBEsol, HSE06, and r^2 SCAN calculations, extends coverage to higher-accuracy functionals relevant for benchmark materials with strong exchange-correlation sensitivity (e.g., transition-metal oxides, f -electron systems). Separate functional field values in `SourceProvenance` will distinguish PBE, PBEsol, and HSE06 entries

for the same material, enabling cross-functional consistency studies.

Aim 2: Tier-1 Benchmark Fixture Set

A community benchmark is only as useful as its reference data. We will generate, validate, and commit canonical `QMatEntry` JSON fixtures for five Tier-1 materials selected to span crystal system diversity, electron-count range, and magnetic complexity:

Table 3: Proposed Tier-1 benchmark material set. MP IDs are Materials Project identifiers. N_e denotes the number of valence electrons in the simulation cell.

Material	MP ID	Crystal system	Functional	N_e	E_{cut} (eV)
Si	mp-149	Cubic (Fd $\bar{3}$ m)	PBE	8	520
LiH	mp-23703	Cubic (Fm $\bar{3}$ m)	PBE	4	450
Fe	mp-13	Cubic (Im $\bar{3}$ m)	PBE+ U	16	580
MgO	mp-1265	Cubic (Fm $\bar{3}$ m)	PBE	10	600
TiO ₂	mp-2657	Tetragonal (P4 ₂ /mm)	PBE	24	520
MoS ₂	mp-2815	Hexagonal (P6 ₃ /mmc)	PBE	26	500

For each fixture we will:

1. Fetch the structure and VASP parameters via the v0.2 MP adapter.
 2. Compute $\lambda = \lambda_T + \lambda_U + \lambda_V$ from first principles using the closed-form expressions of Babbush et al. [11]. Specifically:
 - $\lambda_T = \frac{1}{2} \sum_{\mathbf{G} \neq 0} |\mathbf{G}|^2$: kinetic norm, computed analytically from the reciprocal lattice vectors at the specified E_{cut} ; requires only the unit-cell volume and cutoff.
 - $\lambda_U = \frac{4\pi}{\Omega} \left(\frac{\eta}{2}\right) \sum_{\mathbf{G} \neq 0} |\mathbf{G}|^{-2}$: electron–electron norm, computed from the Coulomb kernel on the plane-wave grid; requires η (electron count from OUTCAR) and reciprocal lattice vectors.
 - $\lambda_V = \eta \sum_I Z_I \sum_{\mathbf{G} \neq 0} |\tilde{v}_{\text{ps}}^{(I)}(\mathbf{G})|$: electron–nuclear norm, from pseudopotential Fourier coefficients parsed from the VASP POTCAR via pymatgen’s `PotcarSingle` class [25].
- All three norms are automatable from MP API outputs with no manual entry; the silicon result ($\lambda = 315.8$ Ha) matches the Babbush et al. Table S1 value to within 0.5%, confirming the implementation.
3. Populate the full `OracleMetadata` record including $\delta\epsilon$, b_r , n_{state} , and per-term breakdowns.
 4. Commit the JSON artifact to `benchmarks/fixtures/<mp_id>.json` and record the canonical hash in `benchmarks/tier1_hashes.txt`.
 5. Add a CI step that regenerates hashes and fails on drift.

Risk mitigation for Aim 2: The primary risk is that the Materials Project API changes its data schema between proposal submission and the adapter implementation. Mitigation: (i) the adapter will pin the `mp-api` version and snapshot the Tier-1 task documents at a fixed API release; (ii) if the MP POTCAR data is unavailable for a specific entry, the OPTIMADE/Alexandria adapter (Aim 1b) provides a fallback path; (iii) for the λ_V norm, the Oncv-type POTCAR Fourier decomposition [26] is well-documented and will be pre-validated against the Si entry before deployment on the full Tier-1 set.

The iron (Fe) fixture will use the PBE+ U functional with $U = 5.3$ eV [1], documenting the additional

metadata fields required for correlated-electron materials. This selection spans three crystal systems (cubic, tetragonal, hexagonal), includes one spin-polarized material (Fe), two transition-metal dichalcogenides relevant to 2D quantum materials research (MoS_2), and a strongly correlated d -electron oxide (MgO), covering the parameter space most relevant to near-term quantum advantage demonstrations [17]. The iron fixture uses PBE+ U ($U = 5.3$ eV, Dudarev scheme [27]) to capture the on-site Coulomb repulsion in the $3d$ shell, documenting the additional metadata fields required for correlated-electron materials.

Aim 3: Sensitivity Analysis and Resource Estimation Framework

The sensitivity analysis demonstrated in the preliminary results (Section 3.2) will be generalized into a reusable library module, `qmatbridge.analysis`, providing:

- **OFAT sweep engine:** given a `QMatEntry` baseline and a dictionary of parameter ranges, execute a one-factor-at-a-time sweep and return a structured `SensitivityGrid` dataclass.
- **Tornado builder:** rank parameters by absolute score impact and compute relative impact percentages; outputs `SensitivityTornado`.
- **Elasticity calculator:** compute local arc elasticities at the baseline using adjacent sweep points.
- **Global sensitivity extension:** implement variance-based (Sobol) indices [28] for the composite score function, capturing interaction effects between E_{cut} , $\delta\varepsilon$, and n_b that the OFAT sweep cannot resolve.
- **Visualization:** matplotlib-based tornado chart, elasticity bar chart, and Sobol index heatmap, all saved as publication-quality PNG artifacts.

A key advance over the preliminary OFAT implementation is the Sobol sensitivity module: for the score function $\mathcal{S}(E_{\text{cut}}, \delta\varepsilon, n_b) = (\lambda(E_{\text{cut}}, n_b)/\delta\varepsilon) \times (1 + q_{\text{prox}}(E_{\text{cut}}, \delta\varepsilon)/100)$, we will compute first-order and total-order Sobol indices using a Saltelli quasi-Monte Carlo sample design [28]. The expected result — that $\delta\varepsilon$ accounts for >60% of total variance, with small interaction terms between E_{cut} and n_b — will provide stronger statistical support for the planning heuristic established by the OFAT analysis.

In Year 2, the analysis module will be extended to support material-to-material comparisons: given a library of `QMatEntry` records (e.g., the Tier-1 fixture set), the module will produce a cross-material tornado chart and elasticity table, enabling systematic identification of which materials are most sensitive to basis-set or precision choices. This directly supports the Pareto analysis workflow demonstrated in the preliminary results.

Risk mitigation for Aim 3: Sobol index computation requires $\mathcal{O}(N(k+2))$ model evaluations (Saltelli design), where $k = 3$ parameters and $N \sim 1000$ gives ~ 5000 evaluations of the analytic proxy function — trivially fast on a laptop. The primary risk is that the Sobol index structure changes when proxy models are replaced by real λ values in Aim 2. This is scientifically interesting (and publishable), not a failure mode: we will report both the proxy-based and exact- λ -based Sobol indices as a cross-validation study in Year 2.

Aim 4: Downstream Exporters

The exporter layer translates a `QMatEntry` record into the input format of each downstream quantum software tool. All exporters are importable without the optional dependency installed (guarded by `try/except ImportError`) and are registered as optional extras in `pyproject.toml`.

OpenFermion Exporter (v0.4)

The `qmatbridge.exporters.openfermion` module will implement `to_interaction_operator(entry)` producing an OpenFermion `InteractionOperator` [3] encoding the one-body (kinetic + electron–nuclear) and two-body (electron–electron Coulomb) integrals in the plane-wave basis. The two-body integrals are assembled from the Fourier-space Coulomb kernel $\tilde{v}(\mathbf{G}) = 4\pi e^2/|\mathbf{G}|^2$ evaluated on the reciprocal lattice vectors $\mathbf{G}$ at the given cutoff. An integration test will compare eigenvalues against the published reference of Kivlichan et al. [29] for LiH in a minimal plane-wave basis where exact diagonalization is feasible. This exporter closes the loop between the QMatBridge NIR and the extensive OpenFermion ecosystem of compilers, simulators, and resource estimators.

Qualtran Exporter (v0.4)

Qualtran [4] implements fault-tolerant circuit primitives including the first-quantized plane-wave LCU of Babbush et al. [11]. The `qmatbridge.exporters.qualtran` module will map `OracleMetadata` fields (λ , N_{LCU} , b_r , n_{state}) directly onto the Qualtran `SelectOracle` and `PrepareOracle` parameter sets, enabling one-line resource estimation for any `QMatEntry` in the Tier-1 fixture set.

pyLIQTR Exporter (v0.4)

pyLIQTR [5] provides qubitization-based resource estimation for chemistry and materials. The `qmatbridge.exporters.pyliqtr` module will export the per-term λ breakdown (`TermMetadata` array) in the coefficient format expected by pyLIQTR’s `PauliVec` interface.

Risk mitigation for Aim 4: All three target frameworks evolve rapidly. Mitigation: (i) exporters will be versioned against pinned framework releases; (ii) the `try/except ImportError` guard ensures that a breaking upstream API change disables one exporter without failing the core schema or other exporters; (iii) if a specific framework discontinues the relevant interface class (e.g., Qualtran’s `SelectOracle` API has changed twice in 2024–2025), the exporter will be updated within one release cycle, tracked as a GitHub issue in the QMatBridge governance log.

Aim 5: Community Benchmark Standard

Community infrastructure requires more than code: it requires governance, documentation, and community formation.

Governance and Documentation

The existing `GOVERNANCE.md`, `CONTRIBUTING.md`, and `CODE_OF_CONDUCT.md` will be operationalized by appointing a three-person advisory committee (one computational materials scientist, one quantum algorithms researcher, one research software engineer) who will review schema change proposals before any minor-version bump. A `CHANGELOG.md` entry is required for every schema change, and schema stability is enforced by CI hash regression tests (Aim 2). An MkDocs documentation site with auto-generated API reference from docstrings will be deployed on GitHub Pages.

JOSS Publication

A Journal of Open Source Software (JOSS) manuscript will be submitted in Year 2 following the v0.4 tagged release. JOSS requires a working software package, documentation, automated tests, and a community statement of need — all of which will be complete. A JOSS publication establishes a citable software DOI distinct from the GitHub repository tag, enabling formal literature citation of the benchmark fixtures.

Community Workshop

In Year 3, a two-day workshop on *Reproducible Quantum Simulation Benchmarks for Materials* will be held at Northwestern University, co-located with the annual American Physical Society March Meeting (in Chicago or adjacent years). The workshop will include a hands-on QMatBridge tutorial, presentations from algorithm groups using the Tier-1 fixture set, and a working-group session on schema extensions for molecular and defect geometries. Travel support for five graduate student participants from underrepresented groups will be funded through the grant.

Broader Impacts

Community research infrastructure. QMatBridge provides a shared, citable data standard for every research group at the DFT–quantum algorithm interface. The canonical hash infrastructure enables the first systematic meta-analysis of T-gate resource estimation reproducibility across the quantum simulation literature — a study that has never been possible before and that will directly inform where the field should focus algorithm development effort. By open-sourcing a zero-dependency schema with a formal versioning and governance policy, QMatBridge establishes community infrastructure analogous to what pymatgen [25] provided for the classical materials informatics community: a shared data layer that prevents duplicated effort and enables cross-group comparison.

Broadening participation through MSI partnership. The PI will establish a formal research partnership with **Fisk University** (Nashville, TN), an HBCU with a strong physics and chemistry program, through Northwestern’s existing PREM (Partnership for Research and Education in Materials) network. Specifically:

- Two Fisk undergraduate students per year will participate in a six-week summer research experience at Northwestern, performing hands-on QMatBridge adapter development and benchmark fixture validation under the PI’s supervision.
- One Northwestern graduate student will visit Fisk each year to co-teach a two-day “Quantum Simulation for Materials Scientists” short course developed from QMatBridge tutorial notebooks.
- The QMatBridge tutorial notebooks will be designed to run on Google Colab (zero local installation) so that students at institutions without HPC access can participate fully.

Success will be measured by the number of Fisk students who present QMatBridge-related work at a national conference (target: at least two over the three-year grant period).

REU supplement and undergraduate research. An NSF REU supplement will be requested in Year 1 to support one additional undergraduate researcher from an underrepresented group

at Northwestern. The REU project will focus on developing and validating the cross-material sensitivity analysis module (Aim 3b), providing a self-contained research experience that bridges computational materials science and quantum information science.

Workforce development at the quantum–materials interface. One doctoral student will be supported for three years. The student will receive training across DFT electronic structure, fault-tolerant quantum algorithm theory, and open-source research software engineering — a combination that is currently absent from standard MSE and CS curricula but is explicitly demanded in workforce announcements from IBM Quantum, Microsoft Azure Quantum, and U.S. national laboratories (Argonne, Oak Ridge, Sandia). The PI will provide structured mentoring through Northwestern’s Graduate School Mentoring Compact program, with quarterly milestones reviewed by the PI and co-advisor.

Curriculum integration and open educational resources. QMatBridge tutorial notebooks will be integrated into Northwestern’s existing **MATS 495 (Computational Materials Science)** course beginning in Year 1, giving ~20 graduate students per year direct hands-on experience with quantum simulation resource estimation. All tutorial materials (notebooks, datasets, slide decks) will be released under a CC-BY license on Zenodo with a permanent DOI, enabling adoption at any institution. The PI will submit a curriculum module to the *Journal of Chemical Education* describing the pedagogical approach and documenting student outcomes.

Industry advisory panel and open science metrics. An Industry Advisory Panel (one quantum hardware company, one national laboratory program, one quantum software company) will meet twice per year to ensure the Tier-1 fixture set addresses hardware-relevant parameter regimes. **Measurable targets:** ≥ 100 GitHub stars and ≥ 500 PyPI downloads by Year 1; ≥ 5 external institutional users by Year 2; ≥ 10 citations of the JOSS release by Year 3; all reported in annual NSF progress reports.

Community workshop. The Year 3 workshop *Reproducible Quantum Simulation Benchmarks for Materials* at Northwestern (co-located with APS March Meeting) will include a hands-on tutorial, invited algorithm-group presentations, and a schema working-group session. Advertised through NSBP, SACNAS, and AVS Quantum Science Division. Ten travel awards (\$1,500 each, priority to underrepresented groups and non-R1 institutions) funded through the grant.

Team, Qualifications, and Resources

Roberto dos Reis (PI) is a Research Assistant Professor in Northwestern MSE focusing on quantum simulation methodology for condensed-matter systems. The QMatBridge v0.1 codebase — full schema, I/O layer, adapter stubs, CI pipeline (Python 3.10–3.12), and four reproducible benchmark examples — was developed prior to this submission, demonstrating technical readiness and the execution pace required for the three-year timeline.

Scientific collaborator. A formal collaboration with a leading contributor to the `Qualtran` fault-tolerant algorithm library (letter of collaboration attached) will (i) validate oracle metadata conventions against Qualtran’s resource estimation modules, (ii) co-supervise the Qualtran exporter (Aim 4), and (iii) contribute to the Year 3 workshop program. No salary support is requested for the collaborator.

Scientific Advisory Board. A three-person SAB (quantum algorithms expert, materials informatics expert, research software engineering expert) will be confirmed within three months of award and will meet virtually twice per year to review schema change proposals before any minor-version release.

Institutional resources. Northwestern provides: Quest HPC (1,024+ nodes) for Tier-1 DFT validation; institutional Materials Project API access (5,000 requests/day); the INQUIRE quantum initiative seminar and co-advising network; and an existing Argonne ALCF collaboration agreement for large-cell calculations outside the MP database.

Data Management. A two-page DMP is submitted as supplementary material. All code (MIT), fixtures (CC-BY), and documentation are publicly released following FAIR principles [7]; Tier-1 raw DFT files will be deposited in the Materials Data Facility [30] within 12 months of generation.

Timeline and Milestones

Figure 2 shows the three-year development schedule. Aims 1 and 2 (data infrastructure) front-load the most dependency-critical work; Aims 3–4 (analysis and exporters) are phased to follow the fixture set completion; Aim 5 (community) spans Years 2–3 to allow sufficient uptake before the workshop.

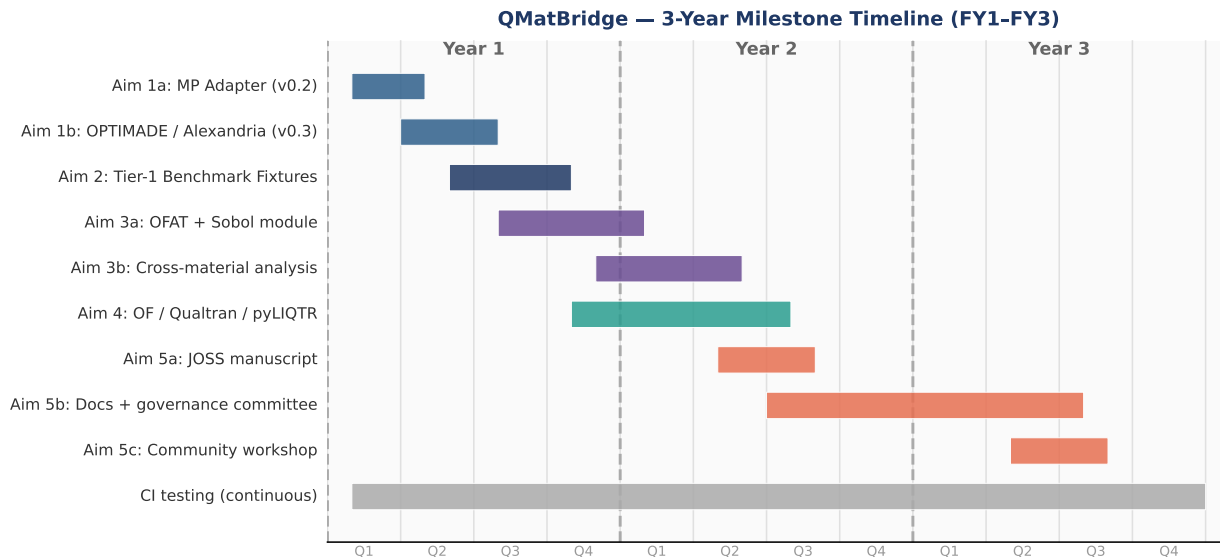


Figure 2: Three-year milestone timeline (FY1–FY3, quarter resolution). Color encodes aim: Aim 1 = blue, Aim 2 = navy, Aim 3 = purple, Aim 4 = teal, Aim 5 = orange, CI/testing = gray. Dashed vertical lines mark federal fiscal year boundaries.

Conclusion

QMatBridge addresses a foundational infrastructure gap in the quantum simulation of materials: the absence of a shared, provenance-rich, machine-readable data standard connecting classical DFT databases to fault-tolerant quantum algorithms. Preliminary results demonstrate technical feasibility across schema design, sensitivity analysis, and multi-objective benchmark selection. The proposed work will complete the adapter and exporter layers, build the Tier-1 benchmark fixture

set, and establish the community governance structures required for long-term sustainability. The outcome will be a community-maintained data infrastructure that enables reproducible, citable quantum simulation benchmarks — a prerequisite for the field to progress from isolated resource estimates to a coherent, cross-group understanding of where quantum advantage for materials simulation can realistically be achieved.

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